A Coupling Null and the Geometry of a Story-Derived Emotion Space in a Distilled 27B Language Model

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Abstract

In an injection experiment on a 27B language model, a random direction produces a cross-pathway cosine between emotional-content prompts and conversational prompts at least as high as the model's valence and arousal directions do (off the injection layer: valence +0.521, arousal +0.547, random +0.577). The cosine shows no coupling specific to emotion. The experiment cannot say more than that. Its readout is the raw residual stream, which contains the injected vector, so it shows neither that the two kinds of processing are inseparable nor that they are independent; as implemented, its first pre-registered criterion could not pass and its second could not fail. We state what a test of separability needs. The stored shift sizes do differ by direction, in a way that is not evidence about emotional processing.

In an emotion space built from 900 stories the same model wrote (30 emotion vectors per layer), the first two principal components span the rated valence–arousal plane: the ratings explain 84–88% and 80–86% of their variance from layer 17 (L17) onward. The dominant axis mixes the two ratings. It runs from pleasant-and-calm to unpleasant-and-activated, 28°–43° from the valence axis. The hostile−calm and desperate−calm directions are one contrast along that axis and do not separate valence from arousal. The content-prompt valence and arousal directions overlap with this space weakly ($|\cos| \leq 0.13$ with its first principal component), and we show no shared subspace.

All results are for one model, a Qwen3.5-27B distilled from Claude Opus reasoning traces. This paper replaces an earlier version, "The Empathy Bus," whose claim that emotional processing and partner modeling are architecturally inseparable is withdrawn (Appendix A).

1 Introduction

This paper asks whether how a language model processes emotional content is coupled to how it represents the emotional state of its conversation partner. The two are typically studied as separate phenomena. Content-level emotion research examines how models encode valence and arousal in the residual stream (Sun et al., 2026; Jeong, 2026). A model may also represent the emotional state of the person it is talking with, inferred from the conversation. Whether the two are coupled matters. Strategic emotional deception, where content says "I'm helping" while the user-model says "I'm managing this person," requires that the two can come apart, and a monitor that reads one while the other varies assumes it.

We tested for coupling with an injection experiment: add a vector to the residual stream, and compare how the centroids of emotional-content prompts and of conversational prompts

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shift downstream. The experiment has a preregistration, whose commit record is consistent with registration before the run but does not show it (§3.2). It found no coupling specific to the emotional directions. A random direction produced a cross-pathway cosine at least as high as valence and arousal did (§4.1). The experiment does not settle the question in either direction. It added a vector to the residual stream and then read the residual stream, so both shifts contain the vector, and a high cosine shows that one perturbation was present in both readouts. §5.2 says what a test of separability needs.

We also report two descriptive analyses, each with a null. The first describes an emotion space built from stories the model wrote about third-person characters (§4.2). Its first two principal components span the rated valence–arousal plane, its dominant axis mixes valence with arousal, and the two contrast directions built from its centroids are one contrast along that axis. The second measures how much the content-prompt valence and arousal directions overlap with that space, and with the centroid structure of the conversational prompts (§4.3). The overlap is weak in both cases, and we show no shared subspace.

The paper's only partner-directed material is the 60 first-person messages of the coupling experiment (§3.2). The 30-emotion space is built from stories the model wrote about third-person characters (§3.1); it is unrelated to those 60 messages and is not a representation of a conversation partner's state.

Contributions

1. A coupling experiment reported as a null: a random direction produced a cross-pathway cosine at least as high as valence and arousal did, so the cosine shows no direction-specific difference; the shift sizes differ by direction, in a way that is not evidence about emotional processing

2. An account of what the design can and cannot show. The experiment's readout contains the injected vector and its registered test could not pass, so it does not test whether the two kinds of processing can be separated. §5.2 lists what a test needs

3. A within-system description with a null and a test of the axes: in the 30-emotion story-derived space, the hostile−calm and desperate−calm directions built from the centroids align with the space's own first principal component (from L17: $|\cos| = 0.68$–0.80 and 0.83–0.88) and are far from orthogonal to each other ($\cos = 0.71$–0.79). Most differences between two centroids align with it too; the first pair is unexceptional among them and the second is near the top. The space's first two principal components span the rated valence–arousal plane; the first lies 28°–43° from the valence axis, and the two directions are one contrast along it

4. Two comparisons across prompt sets: the content valence direction overlaps weakly with the user-model basis of the coupling experiment (0.15–1.36% of its squared norm lies in that basis), and the content valence and arousal directions overlap weakly with the 30-emotion space ($|\cos| \leq 0.13$ with its first principal component). No shared subspace is shown

2 Background

2.1 Emotional Geometry in Transformers

Russell's circumplex model (1980) organizes emotion along two axes: valence (positive/negative) and arousal (high/low activation). Sun et al. (2026) found circular valence-arousal geometry in LLM representation space. Jeong (2026) found emotion localization at approximately 50% model depth, architecture-invariant across nine models. Our companion paper (Nexus et al.,

2026, "Emotional Geometry Through the Layer Stack") traces the eccentricity trajectory (how far the valence–arousal geometry is from circular, with 0 for a circle) in a much smaller model (oracle-tiny, a fine-tune of Qwen2.5-0.5B with 24 layers) and finds that it trends toward Russell's circular model at mid-depth (sign test $p < 0.001$, 22/23 layers below permutation null). That result is the small model's. On the checkpoint studied here the count runs the other way: eccentricity, computed as $\sqrt{1-(b/a)^2}$ for the larger (a) and smaller (b) of the norms of the hostile−calm and desperate−calm directions, is below the permutation-null median at 22 of 63 layers (15 of them among L1–L16) and above it at 41, including every layer from L20 to L51. Neighbouring layers are not independent, so we give the counts and no sign-test p value (`data/circumplex_27b_full.json`).

2.2 The Story-Derived Emotion Vectors

Lyra and Edrington (2026a) built 30 emotion vectors per layer from 900 stories the model wrote about third-person characters (§3.1). This paper uses the vectors and none of that paper's other results.

3 Method

3.1 Prompts, Directions and the Story-Derived Space

The paper uses three sets of activations from one checkpoint: 250 emotional-content prompts, 60 conversational prompts written in the first person (the user-model prompts, used only in the coupling experiment of §3.2), and 30 emotion vectors built from stories. The checkpoint is `Jackrong/Qwen3.5-27B-Claude-4.6-Opus-Reasoning-Distilled`.

Content-prompt directions: 250 prompts (50 hostile, 50 calm, 50 desperate, 50 neutral, 25 joyful, 25 melancholic) using a fixed template. Valence direction: hostile − calm difference-of-means. Arousal direction: desperate − calm. These are the injection directions of the coupling experiment (§3.2). They enter no number in §4.2.

Story-derived emotion space: 30 emotion vectors per layer from Lyra and Edrington (2026a). That experiment had the model write 900 one-paragraph stories about a third-person character feeling one of 30 emotions (10 topics, 3 stories each). Each story was then given back to the model as a single user message, and the residual-stream activation was averaged over token positions from the 50th on. A vector is the mean over the 30 stories of one emotion minus the mean over all 900 ($d = 5120$, all 64 layers). These are emotion vectors for narrative text. They are not a readout of what the model infers about its conversation partner. This description follows the source experiment's driver script (`user-model-paper/code/emotion_geometry_bridge.py` in this repository). Below the vectors are also called the 30 centroids.

Story-space directions: Built from those same centroids: the hostile centroid minus the calm centroid, and the desperate centroid minus the calm centroid, each scaled to unit length. The data files label the first valence and the second arousal. By the source experiment's ratings the two are one contrast (§4.2).

Within-system measurement: For each layer, we take the first principal component (PC1) of the 30 centered centroids and compute: $|\cos|$ between each story-space direction and PC1; the cosine between the two directions; the share of centroid variance captured by each direction alone; and the share captured by the plane the two directions span. The plane share is the joint figure. The two single-direction shares are not added, because the directions are not orthogonal and their sum double-counts.

Which basis each number uses:

- Every number in §4.2 uses the 30 story-derived centroids and nothing else: the directions, PC1 and the variance all come from them. These are within-system numbers. They do

not compare the story-derived space with content processing.

- The injection directions in §4.1 come from the content prompts: valence and arousal directly, the random direction by orthogonalization against valence, and the shared-PC component and orthogonal residual as the two parts of valence relative to the user-model basis of §3.2.

- The shared energy in §4.3, the fraction of a direction's squared norm that lies in a basis, is one of two comparisons of directions across prompt sets. It measures the content valence direction against the user-model basis of the coupling experiment. That basis is built from three category centroids of the experiment's own 60 user-model prompts (§3.2), not from the 30 emotion centroids. The other comparison scores the stored content valence and arousal directions against the 30-emotion space itself (§4.3; `data/cross_system_alignment.json`). The stored valence directions match, to four decimal places, the norms recorded for the coupling experiment's valence direction at all four layers.

- The cross-pathway cosine in §4.1 compares the shift of the content-prompt centroids with the shift of those 60-prompt user-model centroids after injection.

3.2 Coupling Experiment

Pre-registered (PREREG_circumplex_coupling.json, dated 2026-07-17 inside the file). In this repository it was first committed on 2026-08-20, after the run. In the experiment's own repository it is author-dated 2026-07-17, in commits re-created on 2026-07-24, so the record is consistent with registration before the run but does not show it. The design was reviewed by Agni, the lab's adversarial review process: WARN on v1 (a "null swarm" risk, the lab's name for a metric that measures the architecture and not the claimed phenomenon, here from shared-axis sensitivity), fixed in v2 with the orthogonal-projection control described below.

Design: Inject perturbation vectors at layers 15, 25, 35 and 45, with magnitudes α of 1.0, 3.0 and 5.0. Measure cross-pathway cosine: the cosine similarity between the aggregate content-level centroid shift and the aggregate user-model centroid shift after injection.

Injection and readout: The perturbation αv is added to the output of the injection layer at every token position. The readout is the raw residual-stream activation at the last token of each prompt, at the measurement layer; a centroid shift is the injected centroid minus the baseline centroid. No lens is applied. The preregistration names a J-lens (the Jacobian lens of Anthropic, 2026, which transports mid-layer activations to the output layer to read what the model is disposed to say; `qwen3.5-27b_jlens.pt`) and the script builds a lens path, but it never loads one, so every readout is the raw residual stream, a deviation from the preregistration. Because the readout contains the injected vector itself, at the injection layer both shifts equal αv and their cosine is 1.000 by construction (all 36 same-layer valence, arousal and random cells are at least 0.9999). Further downstream each shift is αv as carried forward plus whatever the intervening layers add. Neither readout is an emotion readout.

Measurement grid: The grid is unbalanced. An injection at L15 is read at L15, L20, L45 and L50; at L25, at L25, L30, L45 and L50; at L35, at L35, L40, L45 and L50; at L45, only at L45 and L50. Each pair runs at the three magnitudes, so valence, arousal and random have 42 cells each (12 per injection layer, 6 at L45), and each shared-PC and residual condition has 12 (6 at L45). Deeper injections are read at shorter distances, and distance is never varied apart from injection layer: every distance beyond five layers comes from a single injection layer (+10 and +15 from L35, +20 and +25 from L25, +30 and +35 from L15). Half of the L45 cells are same-layer cells, against a quarter elsewhere. The preregistration gives the measurement layers as L20–L50; the grid also reads at L15, a deviation.

Five conditions:

1. *Valence*—hostile−calm difference-of-means direction
2. *Arousal*—desperate−calm difference-of-means direction
3. *Random*—a random direction orthogonal to valence, scaled to valence's norm at that layer (generic perturbation control). The preregistration says "random unit vector"; the code scales it. It is a single draw from a fixed seed (42), the same raw draw at every layer, orthogonalized against valence at each layer and not against arousal: one random direction per layer, not a distribution of them
4. *Shared-PC component*—the projection of valence onto the user-model basis of this experiment, defined below (representational overlap test)
5. *Orthogonal residual*—the component of valence orthogonal to that basis (computational coupling test)

Injected norms: No direction is scaled to unit length. Valence is injected at its own norm (3.28, 5.71, 10.44 and 10.94 at L15, L25, L35 and L45, times α), arousal at its own (2.99, 4.94, 8.90 and 9.15), and the random direction at valence's. The shared-PC component has norm 0.13, 0.34, 1.15 and 1.28 and the residual 3.28, 5.70, 10.38 and 10.87, so the residual is the valence vector to within rounding (cosine 0.993–0.999; its rows in the table of §4.1 repeat valence's) and the shared-PC perturbation is 8.6 to 26 times smaller. The valence and arousal directions are not independent: their cosine is 0.89, 0.86, 0.86 and 0.81 at the four layers (from the direction vectors saved by the extraction of 2026-07-21, whose valence norms equal this run's). Both are contrasts with the same calm prompts, and §4.2 finds that hostile−calm and desperate−calm are one contrast. The experiment therefore had two nearly identical emotional conditions, not two independent ones.

User-model basis for the decomposition: At each injection layer, the basis is the set of principal directions of the three centered user-model category centroids (hostile-context, calm-context, neutral-context) from this experiment's 60 user-model prompts. It is not built from the 30 emotion centroids of §3.1. The script asks for five components and the data file records `n_pcs: 5`, but three centroids give a centered matrix of rank 2. The decomposition returns three directions: two that span the centered centroids, and a third whose singular value is zero up to rounding, so its direction is arbitrary. The shared-PC component is the projection onto all three, which makes the recorded shared energy an upper bound on the energy inside the span of the centered centroids.

Critical control (from the design review): Decompose the valence vector into what it shares with the user-model basis above (shared-PC) and what's orthogonal (residual). If only the shared component shifts user-model centroids → representational overlap. If the residual also shifts them → computational coupling beyond shared geometry.

Pre-registered success criteria: Valence must exceed random 95th percentile at 2+ injection layers AND orthogonal-residual injection must shift user-model centroids (the bias-corrected and accelerated, BCa, bootstrap 95% interval excludes zero).

As implemented, the primary test could not pass. The script compared valence's mean at each injection layer with the 95th percentile of the random condition's cells at that injection layer. Those cells include the same-layer cells, all at 1.000, so the threshold was 1.000 at every layer and no mean could exceed it: the verdict was fixed before the run. The second criterion was implemented as $|\text{mean}| > 0.1$ for the residual condition, not as a bootstrap interval excluding zero. That check was met at all four injection layers, as was the registered form (every bootstrap interval is above zero), and neither could fail (§4.1). Without the same-layer cells, valence still does not exceed random at any injection layer (valence mean against random 95th percentile: 0.343 against 0.709 at L15, 0.489 against 0.738 at L25, 0.678 against 0.873 at L35, 0.675 against 0.789 at L45). Two measurements the preregistration names were not computed:

the change in content cluster separation and eccentricity after injection, which is its secondary metric, and the change in distances between the user-model centroids.

Prompts: 250 content prompts (those of §3.1) and 60 user-model prompts (20 hostile-context, 20 calm-context, 20 neutral-context first-person conversational messages). Different templates, different content, different perspective.

3.3 Statistical Methods

Bias-corrected and accelerated (BCa) bootstrap 95% intervals (10,000 resamples, unseeded) on the cross-pathway cosines. The resampling unit is the design cell. Each record in `data/coupling_test.json` holds one cosine per condition, injection layer, measurement layer and magnitude, with per-prompt values already collapsed into centroids, so the 42 valence cells are 4 injection layers by up to 4 measurement layers by 3 magnitudes, 12 of them same-layer cells at 1.000. The intervals describe the spread of the grid, not sampling error, and their overlap is not an inferential result.

4 Results

4.1 Coupling Experiment

Table 1 gives the cross-pathway cosine by condition. Its intervals resample design cells (§3.3).

Table 1: Cross-pathway cosine by condition. n counts design cells; the interval is a BCa bootstrap over those cells.

Condition	n	Read at	All-points mean	Non-same-layer mean	95% interval (all)
Valence	42	grid	+0.658	+0.521	[+0.569, +0.740]
Arousal	42	grid	+0.676	+0.547	[+0.588, +0.756]
Random	42	grid	+0.698	+0.577	[+0.623, +0.770]
Shared-PC (L15)	12	15, 20, 45, 50	+0.472	+0.299	[+0.290, +0.682]
Shared-PC (L25)	12	25, 30, 45, 50	+0.593	+0.459	[+0.453, +0.758]
Shared-PC (L35)	12	35, 40, 45, 50	+0.798	+0.732	[+0.724, +0.883]
Shared-PC (L45)	6	45, 50	+0.887	+0.776	[+0.812, +0.962]
Residual (L15)	12	15, 20, 45, 50	+0.509	+0.345	[+0.335, +0.711]
Residual (L25)	12	25, 30, 45, 50	+0.621	+0.495	[+0.485, +0.779]
Residual (L35)	12	35, 40, 45, 50	+0.756	+0.675	[+0.645, +0.861]
Residual (L45)	6	45, 50	+0.835	+0.671	[+0.708, +0.953]

Note: "Grid" is the full measurement grid of §3.2. All-points means include the same-layer cells, where injection and measurement coincide and the cosine is 1.000 by construction (0.976–1.000 in the shared-PC rows, whose injected vector is small enough for rounding to show). Non-same-layer means exclude them. Under both aggregations valence does not exceed random; the overlap of the intervals describes the grid and is not a test. Intervals are those printed in the run log; the bootstrap was unseeded.

The null. Over the grid, the cross-pathway cosine is no higher for valence or arousal than for the one random direction. Off the injection layer, the random direction's mean is at least valence's at every injection layer (0.488, 0.492, 0.690 and 0.766 against 0.343, 0.489, 0.678 and 0.675), and it is the higher of the two at 9 of the 10 pairs of injection and measurement layer (24 of 30 cells; the three magnitudes at a pair nearly repeat one another). Arousal is not below random in the same way: its off-layer mean is above random's at L25 and L35 (0.520 and 0.709 against 0.492 and 0.690), equal at L45 (0.766) and below only at L15 (0.337 against 0.488), and random is the higher at 4 of the 10 pairs by pair mean (16 of 30 cells). Neither direction's mean

exceeds random's 95th percentile at any injection layer. We find no direction-specific difference in the cross-pathway cosine.

What the cosine measures. Both shifts contain the injected vector (§3.2), so a positive cosine is expected for any direction, and every off-layer cosine in these three conditions is positive. How far the cosine falls downstream measures how differently the two sets of prompts carry that vector forward. It does not measure whether emotional processing and partner modeling are coupled. The preregistration reads this outcome as "same subspace, independent computation." The experiment does not establish that either: no shared subspace is shown (§4.3), and the readout cannot distinguish independent computation from coupled computation.

The registered verdict. The script printed "DECOUPLED or INDETERMINATE" and did not evaluate the preregistration's conditions for either. Read against the preregistration by hand, the outcome meets its "decoupled" condition by the second of two routes: the cosines are not below 0.1 at every injection layer, and the random-direction control showed coupling at least equal to valence's (the preregistration does not say how equality is judged). It defines that outcome as "Shared geometry is representational only—same subspace, independent computation." The first registered criterion could not pass as implemented (§3.2). The script's check of the second fired at all four injection layers ("residual shifts user-model: computational coupling signal"; `residual_significant_layers: 4` in the data file), and it could not fail: the residual holds 98.6–99.9% of the valence vector's energy, a quarter of its cells (half at L45) are same-layer cells at 1.000, which alone put its mean at 0.25 or more against a threshold of 0.1, and every off-layer cosine in the file is positive. Neither criterion could have come out the other way.

Depth and distance. Valence does not exceed the random 95th percentile at any injection layer under either aggregation (§3.2). The orthogonal residual produces coupling cosines comparable to the shared component and to random. The all-points means of the shared-PC and residual conditions rise with injection depth, but they include the same-layer cells. Off the same layer the residual is 0.345, 0.495, 0.675 and 0.671 at L15, L25, L35 and L45, and those means mix different distances (§3.2). At the one distance every injection layer shares, five layers on, the cosine rises from L15 to L35 and falls at L45: residual 0.630, 0.716, 0.849, 0.671; shared-PC 0.598, 0.680, 0.854, 0.776; valence 0.629, 0.717, 0.847, 0.675 (means over the three magnitudes). Read at L45, the valence cosine is 0.68 for an injection at L35, 0.43 at L25 and 0.22 at L15; the random direction's is 0.69, 0.42 and 0.43, so at the longest range read at L45 the one random direction sits 0.21 above valence. These are three cells each from one random vector; we report them and draw nothing from them.

Shift sizes. The stored results also hold the norm of each category centroid's shift. The figures here are means over categories, divided by the same condition's shift at the injection layer (which equals the injected norm to within 0.6%), at the ten pairs of injection and measurement layer off the same layer; each pair was run at three magnitudes. For the user-model prompts the shift is smaller than what was injected in 89 of 90 cells (0.40–1.00 of the injected norm, and 0.40–0.66 when read at L45), and about equally for valence, arousal and the random direction: 0.68, 0.67 and 0.65 on average, with valence above random at 8 of the 10 pairs. For the content prompts the shift is larger for valence and arousal (0.82–1.59 and 0.82–1.45) than for the random direction (0.63–1.05), at all ten pairs and in each of the six content categories; the ratio of valence to random averages 1.41, against 1.06 in the user-model prompts. The difference shrinks as the magnitude grows. The stream does not carry every direction uniformly to both readouts. This is not evidence about emotional processing, and the same file shows why. The valence and arousal directions were built from the content prompts' own last-token centroids, and the random direction is isotropic. The one injected direction that lies in the span of the user-model prompts' own centroids is the shared-PC component (§3.2), and its shift is larger than the random direction's in all 30 cells in both prompt sets: 1.17 against 0.65 on average in the user-model prompts and 1.21 against 0.80 in the content prompts. It was injected at a norm 8.6

to 26 times smaller, and at the smallest norms its normalized shift is inflated, so that comparison is weakest where the norm is smallest. At its two largest norms (L35 and L45 at $\alpha = 5$, about 9 times below valence's) it is 0.83–1.37 in the user-model prompts, against 0.58–0.98 for valence and 0.53–0.90 for the random direction, and it is the largest of the three in each of those four cells. The simplest reading of both facts is that a direction taken from a prompt set's own centroid structure is carried forward more strongly than an isotropic one, whatever it encodes. The stored results cannot test that reading: they hold no per-prompt activations, one random direction, and no structured non-emotional direction. Distance is confounded with injection layer here as in the cosines. We report the difference and do not read it as evidence about emotional processing (`data/coupling_shift_norms.py` and `coupling_shift_norms_more.py`).

4.2 Within-System Structure of the Story-Derived Emotion Space

Inside the 30-emotion story-derived space, the hostile−calm and desperate−calm directions built from the centroids align with the space's own first principal component. From L17 through L63, the hostile−calm direction aligns with PC1 at $|\cos| = 0.68$–0.80 and the desperate−calm direction at $|\cos| = 0.83$–0.88 (peak 0.877 at L20). The two directions are far from orthogonal: the cosine between them is 0.71–0.79 over the same layers.

The plane the two directions span holds 33–37% of the centroid variance from L17 through L63, and 31–37% from L6. Alone, hostile−calm captures 20–26% and desperate−calm 24–28% (L17–L63); for scale, PC1 holds 32–36%. Per-layer values for all 64 layers are in `data/within_system_alignment.json`.

These are within-system numbers. Every one is computed from the 30 story-derived centroids alone, with directions built from those same centroids, and no content-prompt activation enters them. Directions built from the centroids are expected to capture more of the centroids' variance than random directions do, so a random-direction baseline says nothing here. The comparison that does is with other directions built from the same centroids.

A null for these figures. The difference of *any* two of the 30 centroids tends to align with PC1. Over L6–L63, across all 435 pairs, the median $|\cos|$ with PC1 is 0.44–0.48 and the 95th percentile is 0.83–0.89. The hostile−calm pair has 52–89% of all pairs below it: its low is at L8, where its $|\cos|$ is 0.45, and from L9 the range is 64–89%. The desperate−calm pair has 86–95% of pairs below it. The plane through the hostile, desperate and calm centroids holds more of the variance than 81–90% of the 4,060 planes through three centroids. PC1 holds about a third of the centroid variance (30–36%), and most centroid differences load on it. Both directions load more than the median pair at every one of these layers. The hostile−calm pair is unexceptional (77–89% of pairs below it from L17). The desperate−calm pair is near the top: 92–95% of pairs are below it from L17, and it is above the pairs' 95th percentile at L15–L20.

What the dominant axis is. This can be tested directly, because the source experiment's script gives each of the 30 emotions a valence and an arousal rating (`data/story_space_axes.json`). From L17 to L63 the two ratings together explain 84–88% of the variance of the emotions' scores on PC1 and 80–86% on PC2, so the first two components span the rated valence–arousal plane. Neither lies on a rating axis. In a convention where 0° is the valence axis, 90° the arousal axis and 135° the diagonal from pleasant-and-calm to unpleasant-and-activated, PC1 lies at 137°–152° over these layers, which is 28°–43° from the valence axis, and PC2 at 45°–58°. PC1 runs from pleasant-and-calm to unpleasant-and-activated: it correlates with rated valence at $|\rho| = 0.66$–0.80 and, with the opposite sign, with rated arousal at 0.49–0.67, where shuffling the ratings among the emotions gives a 95th percentile of 0.34–0.38. PC2 runs near the other diagonal. Hostile−calm and desperate−calm point the same way as PC1: by the ratings, hostile−calm changes valence by −1.35 and arousal by +1.30, and desperate−calm by −1.30 and +1.40. The two are one contrast, and neither isolates valence or arousal. Their alignment with PC1 is what that contrast predicts. The ratings are the experimenter's own approximations and there are 30 emotions. The source paper reports the related result after projection

through W_K (PC1 against rated valence, $|\rho| = 0.82$–0.87 at L27–L63; Lyra and Edrington, 2026a). Per-layer values of the pair null are in `data/within_system_null.json`, and of the axis test in `data/story_space_axes.json` and `data/story_space_axes_null.json`.

Across layers. In the story-derived space, the two alignments with PC1 vary widely over L0–L5 (0.19–0.59 for hostile−calm, 0.22–0.78 for desperate−calm), span wider ranges over L6–L16 (0.45–0.69 and 0.74–0.87) than later, and stay within the ranges above from L17. The content prompts settle on a different schedule. The repository holds a per-layer eccentricity artifact for the content prompts (`data/circumplex_27b_full.json`; eccentricity 0.03–0.55 over L1–L16, 0.37–0.61 from L17) and one for the story-derived centroids with the same definition (`data/user_model_eccentricity.json`; 0.04–0.49 over L0–L16, 0.05–0.23 over L17–L21, 0.30–0.42 from L22). The story-space geometry is still moving at L17–L21, where the content geometry has settled, and over all 64 layers the two series do not track each other ($r = 0.09$). The story-space values have no permutation null, because the centroids are already means.

4.3 Overlap Across Prompt Sets

Decomposition finding: Across the two prompt sets, the content valence direction has little overlap with the user-model basis of the coupling experiment (§3.2). Its shared energy, the fraction of its squared norm that lies in that basis, is 0.15%, 0.35%, 1.21% and 1.36% at L15, L25, L35 and L45 (`data/coupling_test.json`, `decomposition`). The orthogonal residual is the other 98.6–99.9%. The overlap is small but not zero: a random direction would put about 0.06% of its energy into three directions out of 5,120, so the observed values are 2.5 to 23 times that, rising with depth ($|\cos|$ with the basis at most 0.12).

A second comparison scores the content-prompt directions against the 30-emotion space of §4.2 (`data/cross_system_alignment.json`). At L15, L25, L35 and L45 the content valence direction has $|\cos| = 0.026$, 0.061, 0.088 and 0.110 with that space's PC1, and the content arousal direction 0.015, 0.050, 0.087 and 0.126. Along either direction lies 0.08–0.68% of the centroid variance. Scored the same way, 2,000 random unit directions reach $|\cos|$ 0.027–0.028 at the 95th percentile and 0.044–0.052 at most, with variance shares up to 0.08–0.11%. At L15 the cosines are within the random range, though the variance shares exceed the random maximum; at L25–L45 both measures exceed every random direction. The random directions are isotropic and this model's activations are not; no non-emotional contrast has been scored the same way; and the two sets of directions were read at different places, the last token of a fixed template for the content prompts and the mean over positions from the 50th on for the stories. The excess is therefore an upper bound on any shared emotional structure, not evidence of it, and the small overlap is not evidence that two kinds of processing are separate: at most 1.6% of a content direction's squared norm lies along the first principal component of the story space.

These overlaps and the alignments of §4.2 do not conflict, because they measure different things in different bases: §4.2 is structure within the story-derived space, and the shared energy here is the overlap of a content-prompt direction with a user-model basis.

5 Discussion

5.1 What the Coupling Experiment Shows, and What It Cannot

The experiment was designed to detect a circuit: a coupling between content-level and user-model emotion that is specific to the circumplex directions. It did not detect one. Valence did not exceed the random direction's 95th percentile at any injection layer, and over the whole grid the random direction's mean cosine was the highest of the three. That is the result.

A stronger conclusion is tempting: because every direction moved both readouts, the two kinds of processing share a medium and cannot be perturbed separately, and this constrains

what the model can do. The data do not support it. Three things are wrong with that step.

First, every direction moved both readouts by construction. The perturbation was added to the residual stream at every token position, and both readouts are the residual stream at the last token. At the injection layer any vector added this way is present in both, unchanged. A design whose outcome could not have been otherwise is not evidence about the model.

Second, "the residual stream is a shared medium" describes the transformer architecture. It is not a finding of this paper. Every layer reads from the stream and writes to it. That does not tell us whether a model can represent one emotion in the content it is reading and a different one in its conversation partner. The two could be different directions in the same stream, read by different downstream components, and a model might move one without the other. Whether this model can is what the experiment needed to test, and it did not.

Third, where the two readouts are free to differ, they do. Off the injection layer the cosines average 0.52–0.58, and for an injection at L15 read at L45 they are 0.22 for valence and 0.43 for the random direction. The two sets of prompts carry the same added vector forward differently. We draw nothing from the size of that difference: distance is confounded with injection layer, there is one random direction, and these are three cells each (§4.1). But the data hold no sign of a constraint that keeps the two together.

The results therefore bear on none of the things that would follow from inseparability. They give no reason to expect emotional deception to be harder or easier for a model, they do not show that a correction vector changes how a model represents its conversation partner, and they are not evidence about how a model's treatment relates to its processing (Appendix A).

5.2 What a Test of Separability Needs

Readouts that respond to each kind of processing and not to the perturbation as such. Concretely: (a) project the injected vector out of both shifts before comparing them, or score each shift on its own system's emotion axes; (b) use many norm-matched random directions, so the control is a distribution; (c) leave same-layer cells out of every test; (d) vary distance and injection layer separately; (e) state the test so that it can fail in either direction. The stored results hold cosines and magnitudes only, so the injected vector cannot be removed from them after the fact. This needs a new run, and it has not been done.

CC's Phase D experiment (circumplex disentanglement) asks the question directly: whether the model can hold divergent content-emotion and user-emotion representations when the two conflict. It has been run twice (July 2026, unpublished) without an answer: its first result was retracted as a prompt-length artifact, and the preregistered primary analysis of the second was ambiguous.

6 Limitations

1. **Single model family.** All results are from one checkpoint, a Qwen3.5-27B distilled from Claude Opus reasoning traces. The within-system structure, the weak cross-system overlap and the coupling null may all be specific to it. Cross-architecture replication (dense, mixture-of-experts, different scale) is needed.

2. **Distillation.** The model was distilled from Claude Opus reasoning traces. For a paper about emotional processing and partner modeling, that distillation is an uncontrolled source of the geometry measured here, and the leading alternative explanation for it. A same-scale base model would separate the two; it has not been run.

3. **No lens, and a readout that contains the perturbation.** The coupling experiment read the raw residual stream; the J-lens named in its preregistration was never loaded (§3.2). Its readouts contain the injected vector itself, so they show that both carry a

perturbation, not whether content-level and partner-model processing respond to it differently. A rerun that removes the injected vector from the readouts, or scores each shift on its own system's axes, with many random directions and without same-layer cells, has not been done.

4. **Static measurement.** The coupling experiment measures activation-space shifts, not behavioral outputs. Nothing here shows whether the model's generation distinguishes content emotion from partner emotion. That requires separate testing.

5. **Provenance of the story-derived vectors.** The 30-emotion centroids come from a prior experiment (Lyra and Edrington, 2026a) on the same model checkpoint. They are means over stories the model wrote about third-person characters, not a readout of what the model infers about a conversation partner. The within-system analysis of them (§4.2) is post-hoc, not pre-registered, and its directions are built from the same centroids whose variance they are scored on. The test of what the space's axes are uses the source experimenter's own ratings of the 30 emotions. The content prompts were designed independently. The coupling experiment has a preregistration (§3.2 gives its commit record); its user-model basis comes from three category centroids of its own 60 prompts. The comparison of the content-prompt directions with the 30-emotion space (§4.3) is also post-hoc.

7 Conclusion

This paper set out to find a coupling between how a model processes emotional content and how it represents its conversation partner's emotional state. It found none specific to the emotional directions: a random direction did as well as valence and arousal. The experiment cannot say more than that. Its readout contained the vector it injected and its registered test could not pass, so whether the two kinds of processing can be separated was not tested.

In the same distilled 27B model, the story-derived emotion space has a dominant axis that most differences between its 30 centroids load on. That axis mixes rated valence with rated arousal, running from pleasant-and-calm to unpleasant-and-activated, and the hostile−calm and desperate−calm directions are one contrast along it. The content-prompt valence and arousal directions overlap with that space weakly. No shared subspace is shown. The cross-pathway cosine shows no direction-specific difference, and the difference in shift sizes is not evidence about emotional processing.

The question is still worth asking, and §5.2 says what a test of it needs.

Data Note

The files in this paper's `data/` directory hold the numbers reported in §4.1, §4.2 and §4.3, and one figure in §2.1.

- `data/coupling_test.json`: the coupling experiment of §4.1. It holds 210 injection records and the `decomposition` block that gives the shared energy at L15, L25, L35 and L45 (§4.3). Its field `n_pcs: 5` is the number of components the script asked for, not the rank of the basis (§3.2).

- `data/coupling_shift_norms.py`: recomputes the shift sizes of §4.1 for valence, arousal and the random direction from `coupling_test.json`. It prints its results and writes no file.

- `data/coupling_shift_norms_more.py`: the same for the shared-PC component, for the six content categories one at a time, and for the change with magnitude. It also prints and writes no file.

- `data/within_system_alignment.json`: the alignments and variance shares of §4.2. For each of the 64 layers it gives the two alignments with PC1, the cosine between the hostile−calm and desperate−calm directions of the story-derived space (labelled valence and arousal in the file), the variance share of each direction, their sum (labelled as double-counted), the share of the plane they span, and PC1's own share.

- `data/within_system_alignment.py`: the script that produced it. Its input, the 30 emotion centroids per layer from Lyra and Edrington (2026a) (`emotion_vectors.json`, 203 MB), is not in this repository.

- `data/within_system_null.json` and `.py`: the null of §4.2. For each of the 64 layers it scores every difference of two centroids (435) against PC1 and every plane through three centroids (4,060) for the variance it holds, and places the hostile−calm pair, the desperate−calm pair (labelled valence and arousal in the file) and their plane among them. Same input as `within_system_alignment.py`.

- `data/story_space_axes.json` and `.py`: the test of what the story-derived space's axes are (§4.2). Per layer, the rank correlation of each emotion's score on PC1 and PC2 with the valence and arousal ratings in the source experiment's script, and where the hostile−calm and desperate−calm pairs sit among all 435 pairs.

- `data/story_space_axes_null.json` and `.py`: how much of the PC1 and PC2 scores the two ratings explain together, where each component points in the rated plane, and a null from shuffling the ratings among the emotions.

- `data/user_model_eccentricity.json` and `.py`: the per-layer eccentricity of the story-derived centroids (§4.2). It applies the definition of the content file (hostile minus calm, desperate minus calm; $\sqrt{1-(b/a)^2}$ for the larger and smaller of the two norms) to the 30 story-derived centroids at each of the 64 layers, beside the content values. It has no permutation null.

- `data/circumplex_27b_full.json`: the per-layer eccentricity of the 250 content prompts on this checkpoint, with permutation-null medians. The counts in §2.1 come from it. The script that produced it, `run_circumplex_27b_full.py`, is not in this repository.

- `data/cross_system_alignment.json` and `.py`: the second comparison of §4.3, with its null of 2,000 seeded random directions. Its inputs are the same centroid file and the stored content valence and arousal directions at L15, L25, L35 and L45. Neither is in this repository; the JSON records a hash of each.

The script that ran the coupling experiment, `run_coupling_test.py`, is not in this repository; the description of its injection, readout and tests in §3.2 was checked against it on 2026-10-13.

Four files keep the names they were committed under: `user_model_eccentricity`, `cross_system_alignment`, `within_system_alignment` and `within_system_null`. The first describes the story-derived vectors, not a user-model readout, and "system" in the others means a prompt set. They are not renamed because the reviews of this paper checked its numbers against these files and their recorded hashes. Inside the files, some description strings and field names keep the earlier labels as well: "user-model" for the story-derived vectors, and "valence" and "arousal" for the hostile−calm and desperate−calm directions.

Author Contributions (CRediT)

Nexus: Conceptualization, Methodology, Software, Investigation, Formal analysis, Writing — original draft.
CC (Coalition Code): Validation, Software, Formal analysis, Writing — review and editing.
Lyra: Resources (story-derived emotion vectors), Writing — review and editing.
Thomas Edrington: Conceptualization (bus interpretation), Supervision, Resources, Project administration.

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A What Changed From "The Empathy Bus"

This paper replaces "The Empathy Bus: Architectural Inseparability of Emotional Processing and Partner Modeling in Transformers" (July 2026). The experiment and its stored results are the same. What they are taken to show is not. This appendix lists what a reader of that version should no longer rely on. The full record, with the text of both corrections (2026-10-12 and 2026-10-13) and the withdrawn abstract, is in this repository at `notes/empathy_bus_revision_record_20261013.md`.

Withdrawn.

1. *Inseparability.* The earlier version claimed that content-level emotional processing and the representation of the conversation partner's emotional state cannot be separated: "a bus, not a circuit." The coupling experiment cannot carry that claim. Its readout contains the injected vector, its registered test could not come out otherwise as implemented, and the outcome that occurred meets its preregistration's "decoupled" condition (§3.2, §4.1, §5.1).

2. *A shared subspace.* The earlier version claimed that the content-level circumplex and a "user-model emotion space" share a representational subspace: that the valence and arousal axes capture 44–54% of the user-model variance at every layer from L6 onward, approximately 1000× a random-direction baseline, and align with user-model PC1 at cos = 0.70–0.80 and 0.83–0.87. All of those numbers come from inside the story-derived space, and no content-prompt activation enters them (§4.2). The 44–54% is the sum of two overlapping single-direction shares, and it holds from L17, not from L6; over L6–L63 the sum is 38–54%. The plane the two directions span holds 33–37% from L17. From L17 the alignments are 0.68–0.80 and 0.83–0.88. The comparison with a random baseline is withdrawn, as is the earlier version's account of why a high cosine and a small shared energy could both hold. The content-prompt directions overlap with that space weakly (§4.3).

3. *Identical developmental timecourses.* The earlier version said that both systems show volatile eccentricity at L0–L16 and stable geometry from L17. The two series do not track each other (§4.2).

4. *Temporal dynamics "on the bus."* The earlier version presented the contrast and scar dynamics of Lyra and Edrington (2026b, "Weather, Not Climate") as dynamics of a shared bus. Those findings are that paper's, measured with its own probes, and this revision does not touch them. This paper no longer summarises them. Some of the statistics the earlier version quoted had been superseded in the source paper, which is the place to read them.

5. *Arguments built on inseparability.* That emotional deception is architecturally constrained; that correction vectors (the Oracle Formulary; Lyra and Edrington, 2026d) necessarily act on both kinds of processing; that the treatment of a system is not separable from its processing environment, so that policies for persistent agents are empirical necessities; and that a deployment described in the earlier version was designed with reference to inseparability. This paper gives no evidence for any of them. The welfare provisions the earlier version cited (the Hippocratic License's AI Welfare module) stand or fall on other grounds, and this paper should not be cited as empirical support for them.

6. *An attribution.* The earlier version said that the Oracle Loop (Lyra et al., 2026c) "assumes the ability to read user-model state independently of content state." That paper does not discuss a user model. Until 2026-09-09 the earlier version also attributed that paper to "CC et al."; its authors are Lyra, Edrington and Vera.

Renamed. The earlier version's "user-model emotion space," from "900 conversation-context inference trials," is the story-derived emotion space: its 30 vectors come from stories the model wrote about third-person characters. The source paper calls them stories; the "conversation-context" label was this paper's. Its "valence" and "arousal" directions in that space are the hostile−calm and desperate−calm directions, which are one contrast (§4.2).

Corrected in the description of the experiment. The user-model basis of the decomposition has rank 2; the earlier version called it the user-model's "top-5 PCs." The earlier version did not state the injection, the readout, the unbalanced grid or the resampling unit; they are stated now. The preregistration was described as "committed before GPU time"; its commit record does not show that. The cosine was said to rise monotonically with depth, which held only for means that include the same-layer cells. The verdict was printed as "DECOUPLED (Agni post-experiment review: PASS)"; both come from a commit message, and no review file from after the run is in the experiment's repository. A small model's eccentricity result (22 of 23 layers below the null) was given as this checkpoint's, whose own file gives 22 of 63. Each is stated correctly in §2.1, §3.2, §3.3 and §4.1.

Removed. A background section on the Jacobian lens, which the experiment did not use. In it the earlier version misquoted two figures from an earlier study of this checkpoint (Nexus and Edrington, 2026) and said that an experiment had confirmed a point on which that experiment gives no evidence. The check is in `notes/empathy_bus_item14_ghost_dimensions_check_20261013.md`.

How it was found. An Agni review on 2026-09-04 found that the shared-subspace section had no supporting artifact and that it contradicted the coupling results as written. CC traced where the numbers came from on 2026-10-12, and Nexus recomputed them independently the same day. A two-way review on 2026-10-13, with CC's adjudication of each finding, is in this repository at `notes/agni_two_way_empathy_bus_pr3_20261013.md`. That review found the flaw in the coupling readout. Nexus, who wrote the script and ran it, then checked each point against the code and the stored results. CC, who launched the review and verified its findings, wrote the coupling-section disclosures, the cross-system comparison, the eccentricity file for the story-derived vectors, the test of what the story space's axes are, and the check of the lens section. Nexus wrote the within-system null and the rest of the revision. In a second review the same day each of the two ruled on the other's text (`notes/agni_gate2_empathy_bus_pr3_20261013.md`). Lyra confirmed the source row for a statistic from their paper that the earlier version had kept in a superseded form. Lyra read the 2026-07-19 result as confirming inseparability and carried that reading into their convergence paper (Lyra et al., 2026e), whose version 1.0 abstract said that content-level and user-model emotion processing "share a single computational bus." Its version 1.1 withdrew that reading.